# BSAN 400 Introduction to Machine Learning

# Lecture 5. Tree-Based Methods[1]

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[1] Partially based on Hastie, et al. (2009) ESL, and James, et al. (2013) ISLR

# Overview

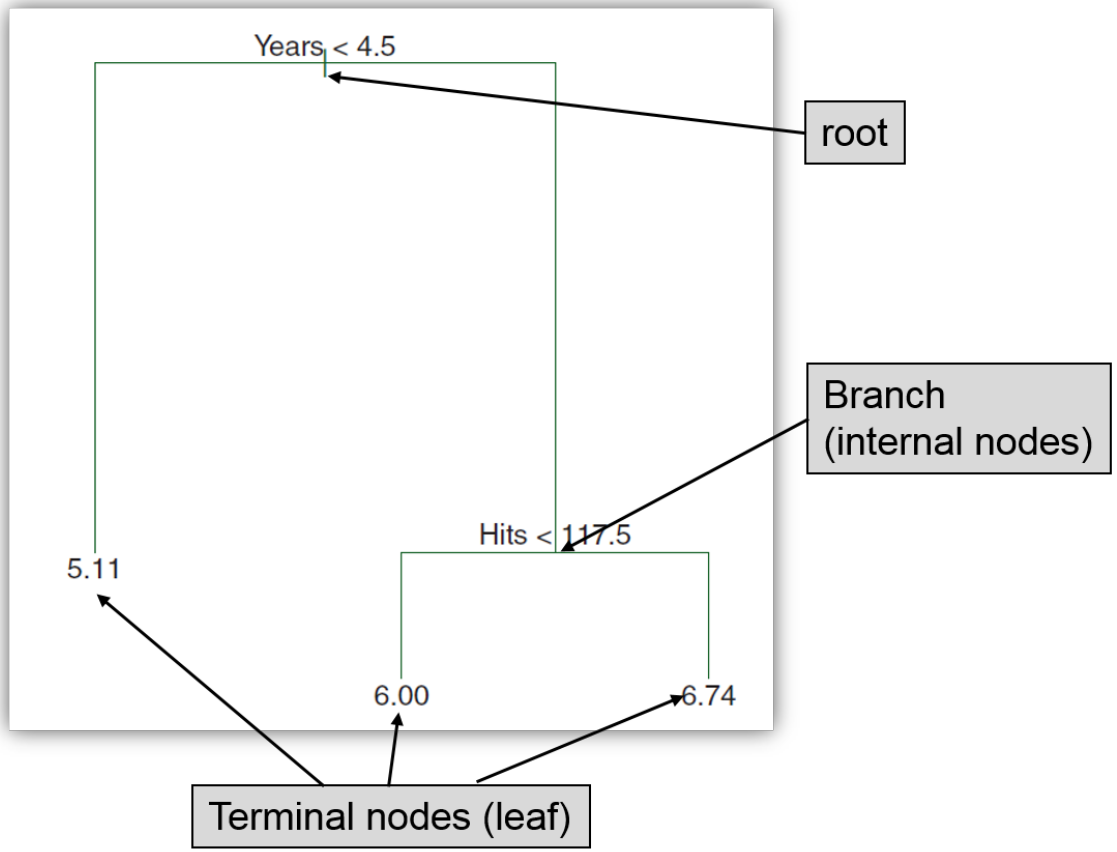
- Supervised learning
- Simple for interpretation
- Nonparametric method

We will talk about

- Classification and Regression Tree (CART)
- Bagging, Random Forest, and Boosting

# A Simple Example of Regression Tree

- Predicting baseball player's log-salary using
  - X1: number of years he has played in the major leagues;
  - X2: number of hits he made in previous year



# Prediction

- The values in terminal nodes
- Top-down
- At each splitting point, go left if TRUE

# Data Partition — Stratification

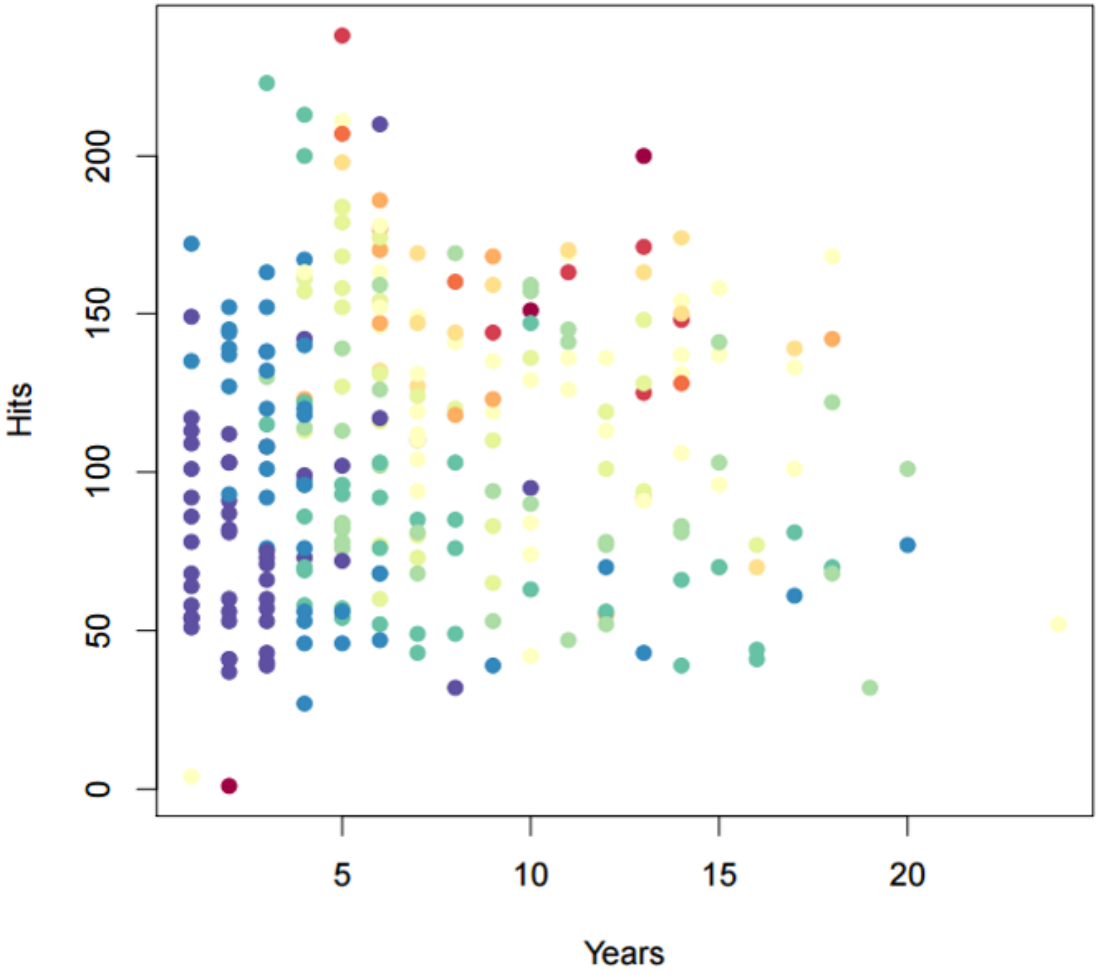
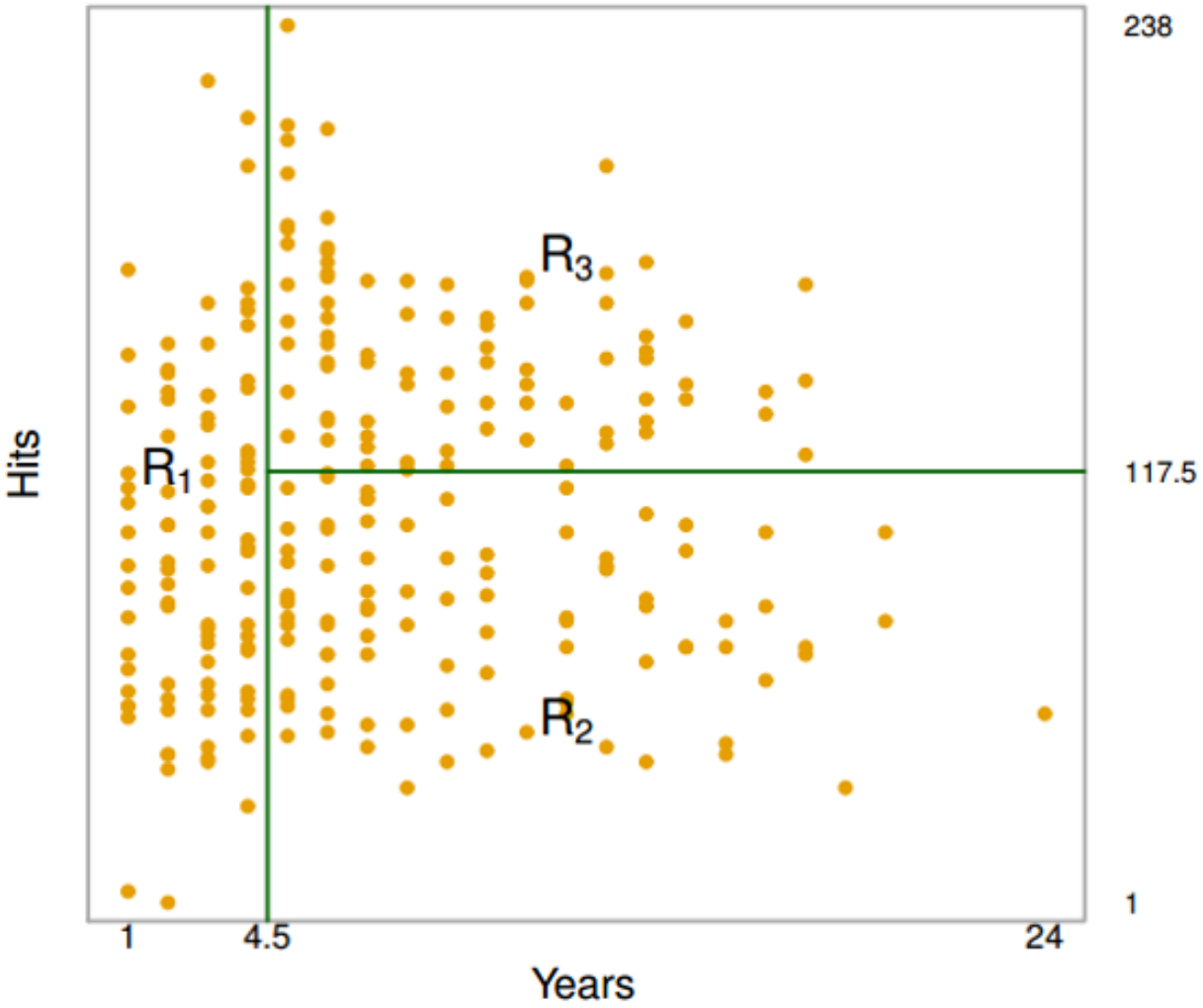



Figure: Salary is color-coded from low (blue, green) to high (yellow, red)

# Data Partition — Stratification



# CART

- Classification and regression tree
- Introduced by Leo Breiman et. al, 1984.
- Binary splitting for variable $j$ at point $s$.
- *Top-down* and *Greedy* algorithm
  - Start from the top of tree
  - The split is optimal at current step

**Question:**

- How to determine the optimal splitting point?
- What criteria should we use?
- When to stop growing the tree?

# Regression Tree

- For each split, we try to determine two regions $R_1$ and $R_2$ by finding the optimal splitting point $X_j = s$, that is

$$R_1 = \{\text{Data}|X_j \leq s\} \quad \text{and} \quad R_2 = \{\text{Data}|X_j > s\}$$

  This is done by scanning all possible splitting point for each $X$.
- Such optimal splitting point minimizes residual sum squares

$$\sum_{i \in R_1} (y_i - \bar{y}_{R_1})^2 + \sum_{i \in R_2} (y_i - \bar{y}_{R_2})^2$$

  where $\bar{y}_{R_1}$ and $\bar{y}_{R_2}$ are average of $Y$ in each region.
- Repeat this process for each region until the decrease of RSS is not significant.

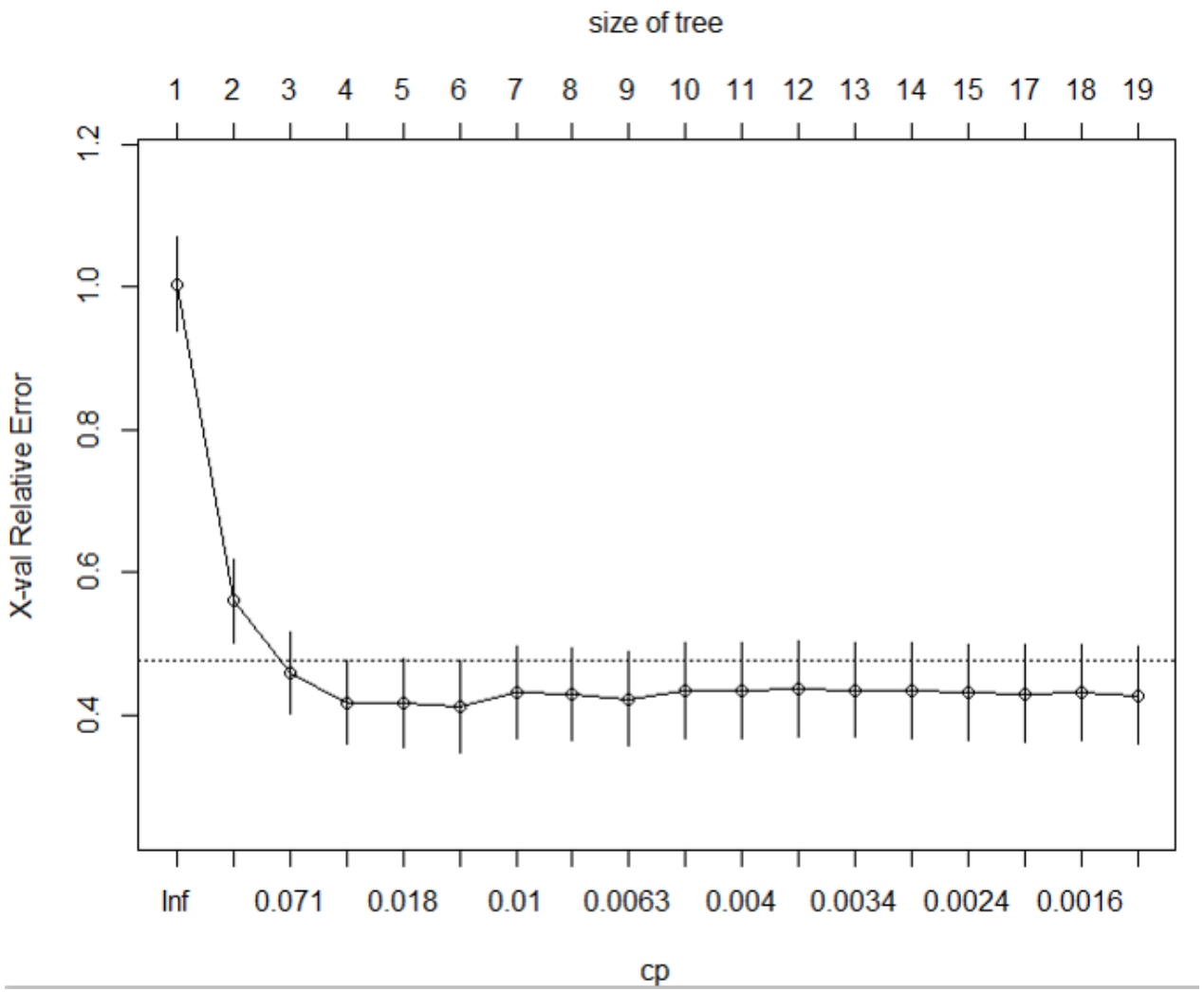
# Stopping Strategy — Cost-Complexity Pruning

- To avoid overfitting!
- Suppose $T_0$ is a very large tree (overfitted tree). Denote $T$ as a subtree of $T_0$, and $|T|$ as the size of that tree (number of terminal nodes). We minimize the cost complexity criterion

$$C_\alpha(T) = \sum_{m=1}^{|T|} RSS_m + \alpha|T|,$$

  where $RSS_m$ is residual sum squares in region (terminal) $m$.
- Larger $\alpha$ penalizes the tree size ($|T|$) more
  - Recall how do we use AIC/BIC for variable selection
  - Every $\alpha$ corresponds to a unique optimal tree $T_\alpha$.
- We use cross-validation to choose the optimal $\alpha$.

# Pruning a Tree in R




- cp is the complexity parameter $\alpha$ we have discussed.

# Classification Tree

- Outcome is categorical variable
- The same idea as for regression tree
- The only difference is the splitting criteria
- What should we use for the criteria?

# Splitting Criteria for Classification Tree

- Misclassification rate
- Gini index

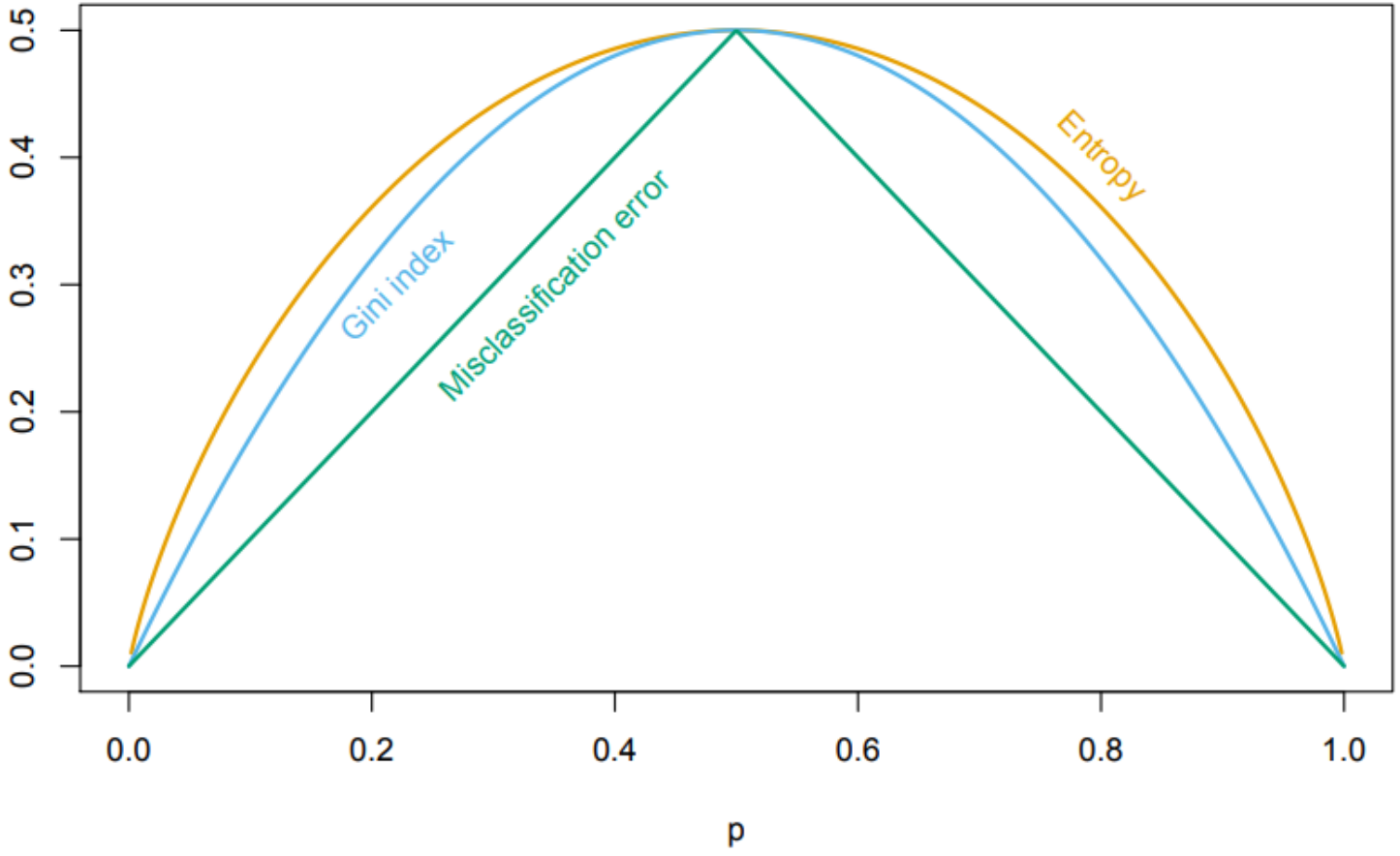
$$\text{Gini index} = \sum_{k \neq k'} \hat{p}_k \hat{p}_{k'} = \sum_{k=1}^{K} \hat{p}_k (1 - \hat{p}_k)$$

  where $\hat{p}_k$ is the proportion of $k$ category observation in one node.
- Entropy

$$\text{Entropy} = -\sum_{k=1}^{K} \hat{p}_k \log \hat{p}_k$$

For each split, gini index (or entropy) is calculated for two regions, and the weighted average is used as the criterion to be *minimized*.

# Illustration



# Example

- Suppose a node contains binary outcome (0 or 1) with each 400 observations.
- Split strategy 1: (300, 100) and (100, 300)
- Split strategy 2: (400, 200) and (0, 200)
- What is the misclassification rate, Gini index, and Entropy for each split strategy?

# Categorical Predictor

- If a categorical predictor has $q$ possible values, how many possible splits?
- Computationally infeasible for large $q$.
- Solution for binary outcome:
  - Sort the $q$ classes according to the proportion of "1" in each of the $q$ classes.
  - Split the ordered variable as if it is numeric variable.
  - This split is the optimal in terms of Gini index and Entropy.
- But we should avoid categorical variable which has too many categories. Why?

# Asymmetric Cost

- Similar idea as we discussed in last lecture
- Define a loss matrix with different weights in misclassified cells.

| | **Pred=1** | **Pred=0** |
|---|---|---|
| **True=1** | 0 | $w_0$ |
| **True=0** | $w_1$ | 0 |

- Loan application example (1=default): $w_1 < w_0$
- The weights are incorporated during each split

# Bagging

- Limitation of a single tree: high variance, unstable
  - Change of points in the sample would lead to different split.
  - Current split heavily relies on previous splits.
- Bagging — Bootstrap aggregation

$$\hat{f}_{\text{bag}}(x) = \frac{1}{B}\sum_{b=1}^{B}\hat{f}_b(x)$$

  - Leo Breiman (1996)
  - Fit many trees on bootstrap samples, and then take average.
  - Variance can be significantly reduced
- For classification tree, how to take average?

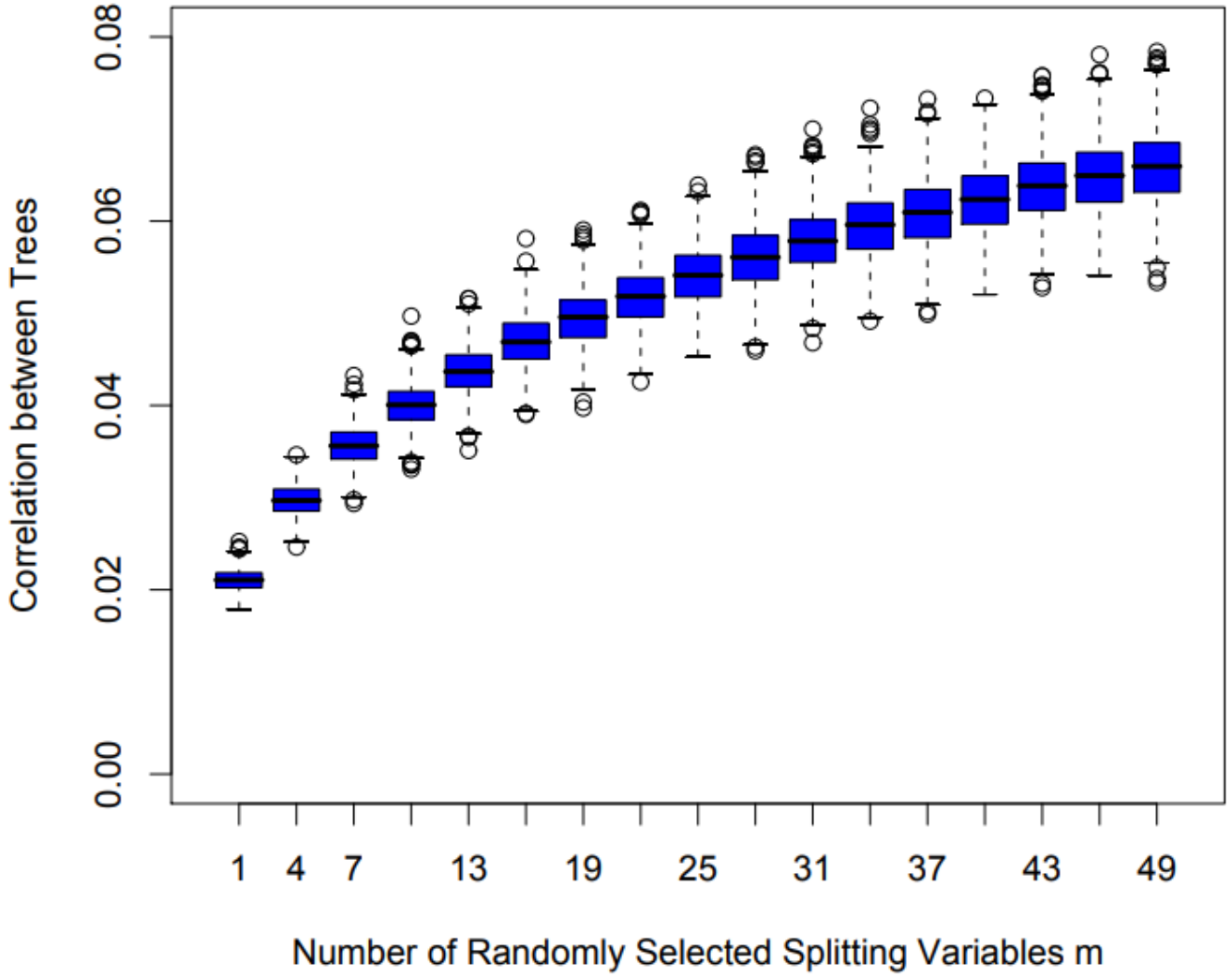
# Out-of-bag (OOB) Prediction

- Similar to cross-validation
- A type of testing error — to prevent overfitting
- For every bootstrapped sample, it is used as training sample, while the rest is used as testing sample. OOB is the averaged error of such testing sample.

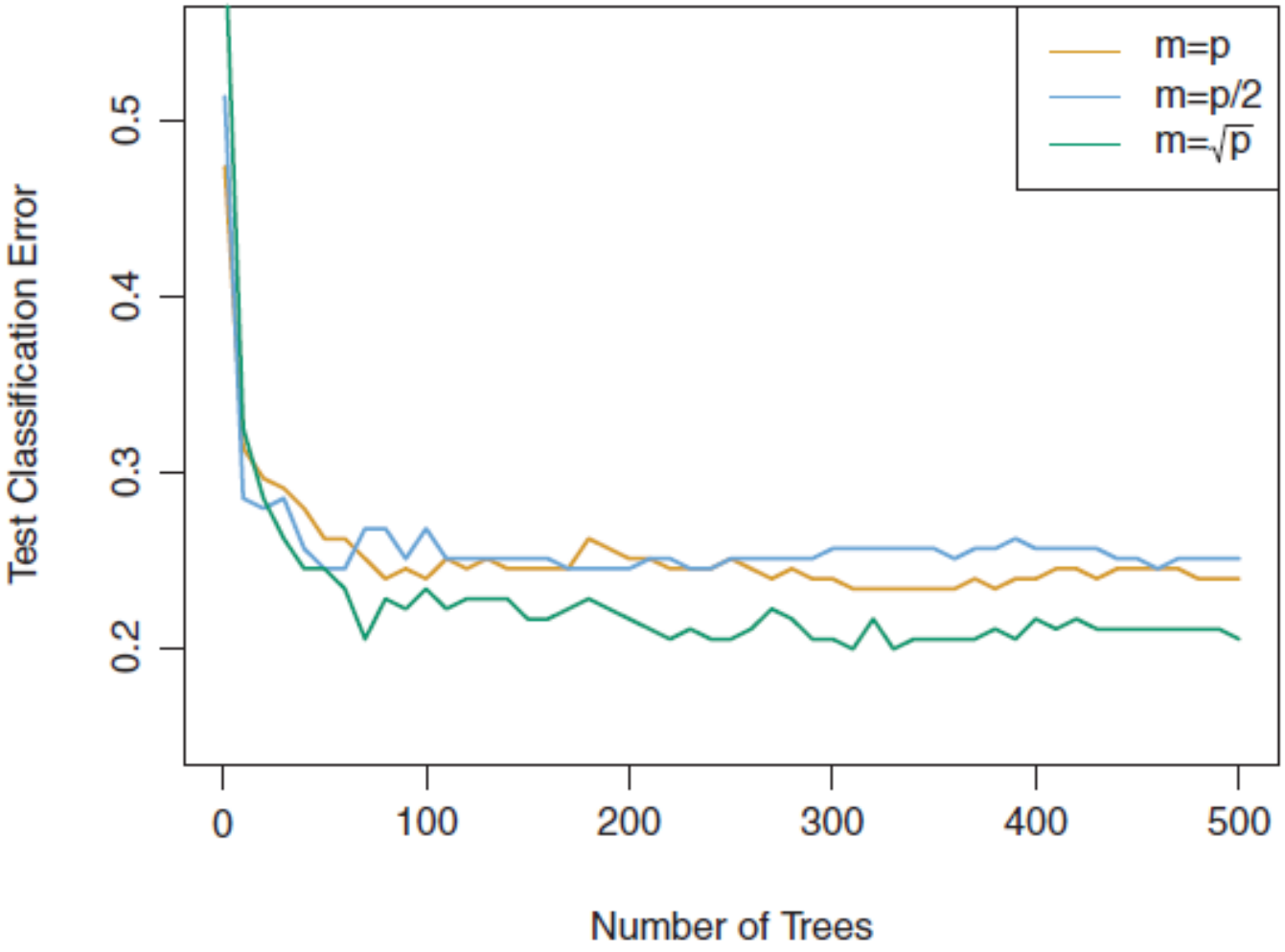
# Random Forests

- Improvement of Bagging
  - The trees in Bagging are correlated to each other
  - It may add variance and prediction error
- Goal: decorrelate a series of trees while maintain strength
- How RF works: randomly choose $m$ predictors as candidate splitting variables for each split
- Choice of $m$
  - Smaller $m$ decreases the correlation, but also decreases strength
  - Larger $m$ increases the correlation, but also increases strength
  - Use OOB error to determine the optimal $m$
  - By default, $m \approx \sqrt{p}$ for classification and $p/3$ for regression
- See the inventor's (Leo Breiman) website for more details.

# Illustration of De-correlation



# Change of Testing Error across $m$



# Gradient Boosting

- Ensemble method
- Based on small trees (weak learner)
- Additive – adding up all small trees
- Pros:
  - Powerful algorithm, high prediction accuracy
  - Small variance (advantage of ensemble methods)
- Cons:
  - Computational expensive for large data
  - Lack of interpretability
  - May overfit data

# Boosting Regression Tree

- Additive: sequentially grow the tree
- Combination of many small trees
- Boosting for regression tree
  1. Set $\hat{f}(x) = 0$, and $r_i = y_i$
  2. For $b = 1, \ldots, B$, repeat:
     - Fit a small tree $\hat{f}^{(b)}(x)$ with $d$ splits based on the training sample
     - Update residual $r_i \leftarrow r_i - \lambda \hat{f}^{(b)}(x)$, and treat this residual as new response, where $\lambda$ is shrinkage parameter
     - Update the tree $\hat{f}(x) \leftarrow \hat{f}(x) + \lambda \hat{f}^{(b)}(x)$
  3. Output the boosted regression tree

$$\hat{f}(x) = \sum_{b=1}^{B} \lambda \hat{f}^{(b)}(x)$$

# Gradient Boosting Machine (GBM)

- A more generalized version.
- User supplied differentiable loss function (e.g. squared loss)

$$\sum_{i=1}^{n} L(y_i, f(x_i)) = \sum_{i=1}^{n} (y_i - f(x_i))^2$$

- The negative **gradient** for each $L(y_i, f(x_i))$ is essentially residual or pseudo residual.
- Then fit a new regression tree to approximate the gradient (fit the residual).
- For binary classification, the loss can be binomial deviance (equivalent to cross entropy, see ESL p346), but the weak learner in each iteration is still a regression tree that approximate the gradient of the deviance $(y_i - p_i)$.

# The algorithm (ESL p387)

**Algorithm 10.4** *Gradient Boosting for $K$-class Classification.*

1. Initialize $f_{k0}(x) = 0,\ k = 1, 2, \ldots, K$.
2. For $m$=1 to $M$:
   (a) Set
   $$p_k(x) = \frac{e^{f_k(x)}}{\sum_{\ell=1}^{K} e^{f_\ell(x)}},\ k = 1, 2, \ldots, K.$$
   (b) For $k = 1$ to $K$:
      i. Compute $r_{ikm} = y_{ik} - p_k(x_i),\ i = 1, 2, \ldots, N$.
      ii. Fit a regression tree to the targets $r_{ikm},\ i = 1, 2, \ldots, N$, giving terminal regions $R_{jkm},\ j = 1, 2, \ldots, J_m$.
      iii. Compute
      $$\gamma_{jkm} = \frac{K-1}{K} \frac{\sum_{x_i \in R_{jkm}} r_{ikm}}{\sum_{x_i \in R_{jkm}} |r_{ikm}|(1 - |r_{ikm}|)},\ j = 1, 2, \ldots, J_m.$$
      iv. Update $f_{km}(x) = f_{k,m-1}(x) + \sum_{j=1}^{J_m} \gamma_{jkm} I(x \in R_{jkm})$.
3. Output $\hat{f}_k(x) = f_{kM}(x),\ k = 1, 2, \ldots, K$.

# AdaBoost — Boosting for Classification

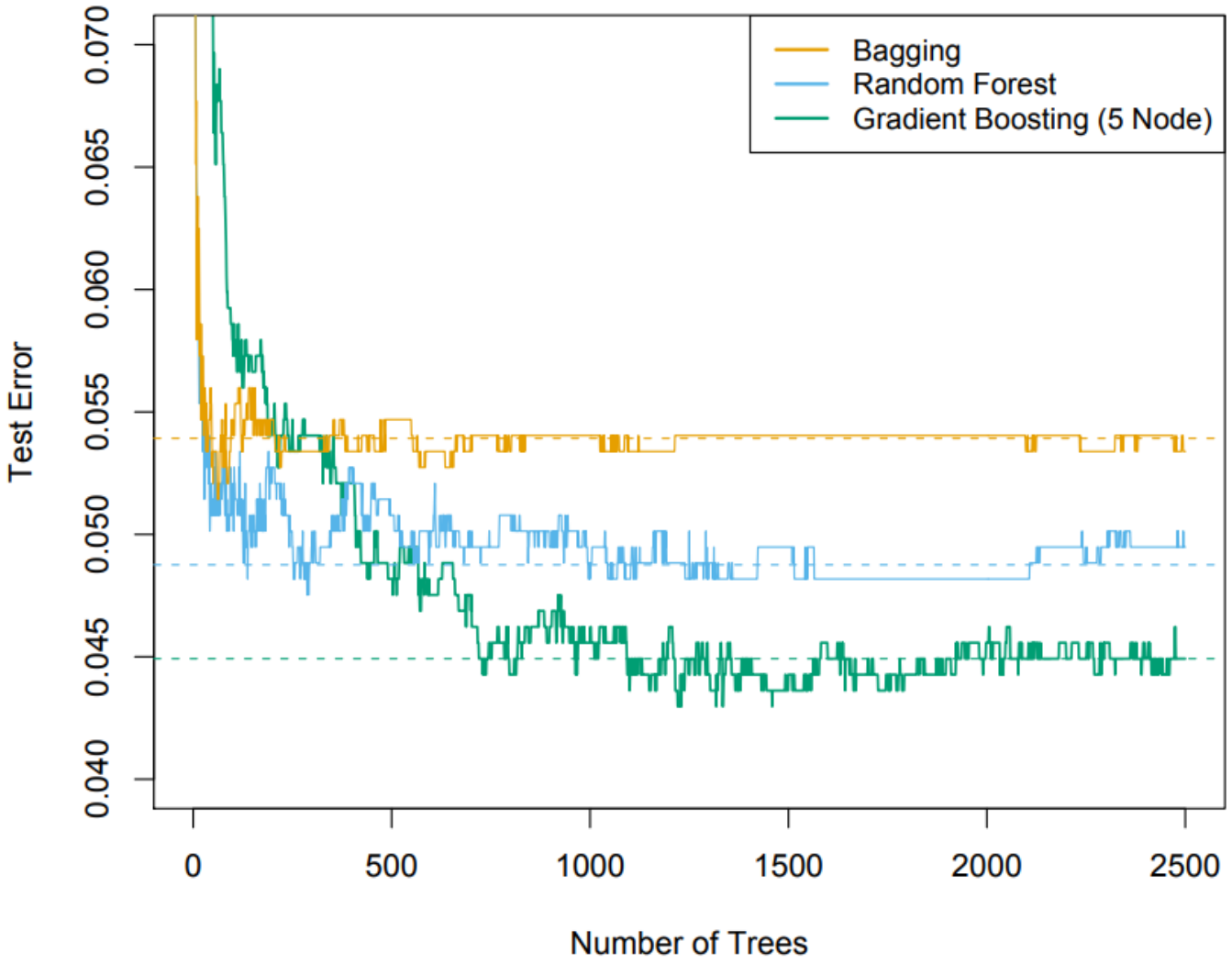
- Originally designed for classification $Y = \{-1, 1\}$
- Algorithm:
  1. Assign equal weights to all observations $w_i = 1/N$
  2. For $b = 1, \ldots, B$, repeat:
     - Fit a classifier $G_b(x)$ to the training sample using weights $w_i$
     - Compute weighted misclassification error

$$\text{err}_b = \frac{\sum_{i=1}^{N} w_i \mathbb{I}(y_i \neq G_b(x_i))}{\sum_{i=1}^{N} w_i}$$

     - Compute $\alpha_b = \log((1 - \text{err}_b)/\text{err}_b)$
     - Update weights $w_i \leftarrow w_i \exp\{\alpha_b \mathbb{I}(y_i \neq G_b(x_i))\}$, and normalize such that the sum equals to 1.
  3. Output $G(x) = \text{sign}\{\sum_{b=1}^{B} \alpha_b G_b(x)\}$.

Here is the StatQuest video that may help you to understand the idea.

# Peformance Comparison



# XGBoost – Extreme Gradient Boosting

- Scalable tree boosting
- Recently developed by Tianqi Chen in 2016.
- One of the best supervised learning algorithms (Machine Learning Challenge Winning Solutions)

# General idea

We omit the mathematical details here. For more detail, please see the official tutorial, and original paper.

- Similar to traditional gradient tree boosting.
- Added regularization which prevents overfitting.
- Optimized the utility of computing power.